Math 4441: Probability and Statistics

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Chapter 2: Random Variables (Part 2)

Bivariate and Multivariate Random Variables

Lectures 09 – 12

Math 4441: Probability and Statistics

References:

1. Goodman and Yates – Introduction to Probability and Stochastic Processes, 3rd Edition
2. Scott Miller and Donald Childers – Probability and Random Processes, 2nd Edition
3. Bertsekas and Tsitsiklis – Introduction to Probability, 2nd Edition

Introduction

- Chapter 2 (Part 1) explained random experiments with outcome as a single numerical value from a discrete and continuous set.
- The probability models were developed for discrete and continuous random variables
- The probability models include
  - Probability mass function (PMF)
  - Probability density function (PDF)
  - Cumulative distribution function (CDF)

- We will study random experiments with two numerical values as outcomes
    - Both from discrete set (Bivariate discrete distribution)
    - Both from continuous set (Bivariate continuous distribution)
    - One from a discrete set and the other from a continuous set (Bivariate mixed distribution)
- For example, in the classification of a set transmitted and received signals, each signal can be classified as – high, medium and low quality
- Random values might represent
    - Number of high-quality signals received
    - Number of low-quality signals received
- Accordingly, we can use random variables X and Y to represent the random values

The simultaneous randomness of X and Y or the joint probability distribution of them will represent the probability model

- Similarly, two random values from two continuous sets might be associated with a single random experiment,
    - Need probability models for simultaneous behavior of two continuous random variables, X and Y.

Finally, a bivariate mixed distribution deals with probability models associated with one discrete and one continuous random variable

Bivariate discrete distributions

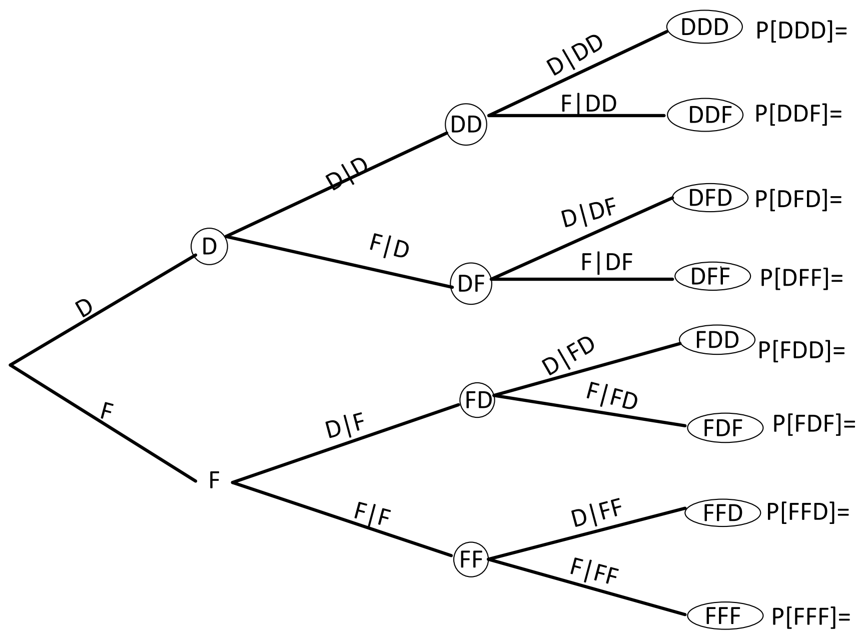
A bivariate discrete distribution is associated for random experiments with

- A pair of discrete random variables
- Joint discrete random variables
- Bivariate discrete random variables

And the probability models include:

- Joint probability mass function (JMPF)
  - Marginal probability mass function (MPMF)
- Joint cumulative distribution function (JCDF)
  - Marginal cumulative distribution function (MCDF)

Example: Send three packets from one computer to another



Sample space of the experiment

$S = \{FFF, FFD, FDF, FDD, DFF, DFD, DDF, DDD\}$

Random variables:

X: # of successful deliveries

$S_X = \{0, 1, 2, 3\}$

Y: # of successful deliveries before any failure occures

$S_Y = \{0, 1, 2, 3\}$

- Each outcome specifies a pair of values of X and Y.
- Single random variable: an outcome represents a point on the number line
- Bivariate random variables: an outcome represents a point on the 2-D plane
- Let $g(s)$ be the function that transform each outcome into a pair of random values. Then

$$g(DDD) = (3,3), \quad g(DDF) = (2,2), \quad g(DFD) = 2,1), \quad g(DFF) = (1,1)$$
$$g(FDD) = (2,0), \quad g(FDF) = (1,0), \quad g(FFD) = (1,0), g(FFF) = (0,0)$$

- Observing the tree diagram, we have

$$P[DDD\} = \frac{27}{64}, P[DDF] = \frac{9}{64}, P[DFD] = \frac{9}{64}, P[DFF] = \frac{3}{64}, P[FDD] = \frac{9}{64},$$
$$P[FDF] = \frac{3}{64}, P[FFD] = \frac{3}{64}, P[FFF] = \frac{1}{64}$$

The probability model of a single discrete random variable represents

- Set of real numbers S_X – each representing an event – a function $g(\cdot)$ converts each outcome into a real number
- Real numbers are points on the number line
- Probability of occurring each point (a number) as the outcome of the experiment – probability of occurring the event $\{X = x\}$ or $\{X \leq x\}$

The probability model of a pair of random variables represents

- Set of pair of real numbers S_{XY} – each representing an event – a function $g(\cdot)$ converts each outcome into a pair of real numbers
- Pair of real numbers are points on a 2-D plane
- Probabilities of occurring each point (a pair of numbers) as the outcome of the experiment – probability of occurring the event
  $\{X = x\} \cap \{Y = y\}$ or $\{X \leq x\} \cap \{Y \leq y\}$

The set of pairs of simultaneous values of X and Y,

a point (x, y) on the 2-D plane,

Can be given by

$S_{XY} = \{(3,3), (2,2), (2,1), (2,0), (1,1), (1,0), (0,0)\}$

The probabilities of the pairs of values (or the points) as the outcomes are:

$P[(3,3)] = P[X = 3, Y = 3] = P[DDD] = 27/64$

$P[(2,2)] = P[X = 2, Y = 2] = P[DDF] = 9/64$

$P[(2,1)] = P[X = 2, Y = 1] = P[DFD] = 9/64$

$P[(2,0)] = P[X = 2, Y = 0] = P[FDD] = 9/64$

$P[(1,1)] = P[X = 1, Y = 1] = P[DFF] = 3/64$

$P[(1,0)] = P[X = 1, Y = 0] = P[FFD] + P[FDF] = 6/64$

$P[0,0] = P[X = 0, Y = 0] = P[FFF] = 1/64$

Joint Probability Mass Function (JPMF)

Definition 4.1 (Joint Probability Mass Function). The joint probability mass function of two discrete random variables X and Y, denoted as $P_{XY}(x, y)$, is defined as

$$P_{XY}(x, y) = P[X = x, Y = y],$$

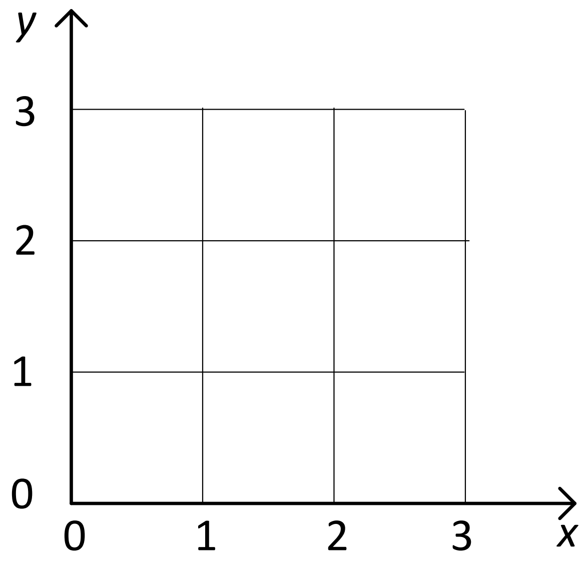
and it satisfies,

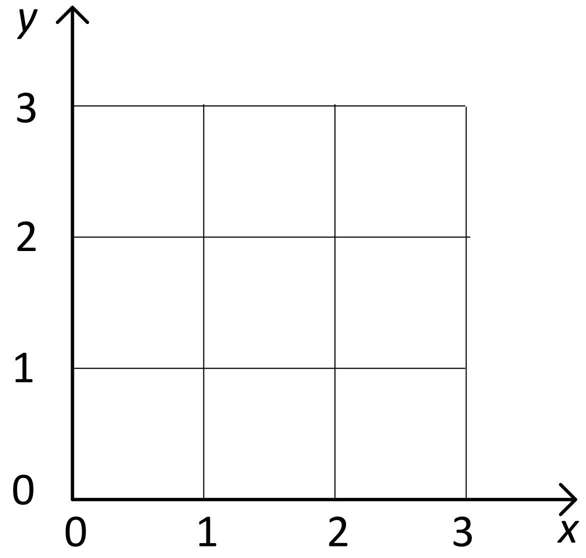
$$\begin{gathered} P_{XY}(x, y) \geq 0; \\ \sum_{x} \sum_{y} P_{XY}(x, y) = 1 \end{gathered}$$

The PMF of a single discrete random variable is represented as a list,

The JPMF of a pair of random variables can be represented in a number of ways.

The bivariate PMF or the JPMF can be represented as a graph





The bivariate PMF or JPMF can be represented as a list as well

$$P_X(x) = \begin{cases} 27/64, & x = 3, y = 3 \\ 9/64\,, & x = 2, y = 2 \\ 9/64, & x = 2, y = 1 \\ 9/64\,, & x = 2, y = 0 \\ 3/64, & x = 1, y = 1 \\ 6/64, & x = 1, y = 0 \\ 1/64\,, & x = 0, y = 0 \\ 0, & \text{otherwise} \end{cases}$$

A third presentation of the joint PMF is the matrix given below:

$P_{XY}(x,y)$	$y = 0$	$y = 1$	$y = 2$	$y = 3$	
$x = 0$					
$x = 1$					
$x = 2$					
$x = 3$					

Summary:

1. Bivariate distributions are used for experiments with two outcomes
2. Random variable X represents one value along the x-axis and Y represents the other value along the y-axis
3. Outcomes are an ordered-pair (x, y) and represents a point on the 2-D plane
4. JPMF of X and Y is defined as
   $P_{XY}(x, y) = P[X = x, Y = y]$
   And represents the probability of simultaneously occurring the value of x and y, i.e., occurring the even $\{X = x\} \cap Y = y\}$
5. Marginal PMFs of X and Y are given by
   $P_X(x) = \sum_y P_{XY}(x, y)$ and $P_Y(y) = \sum_x P_{XY}(x, y)$

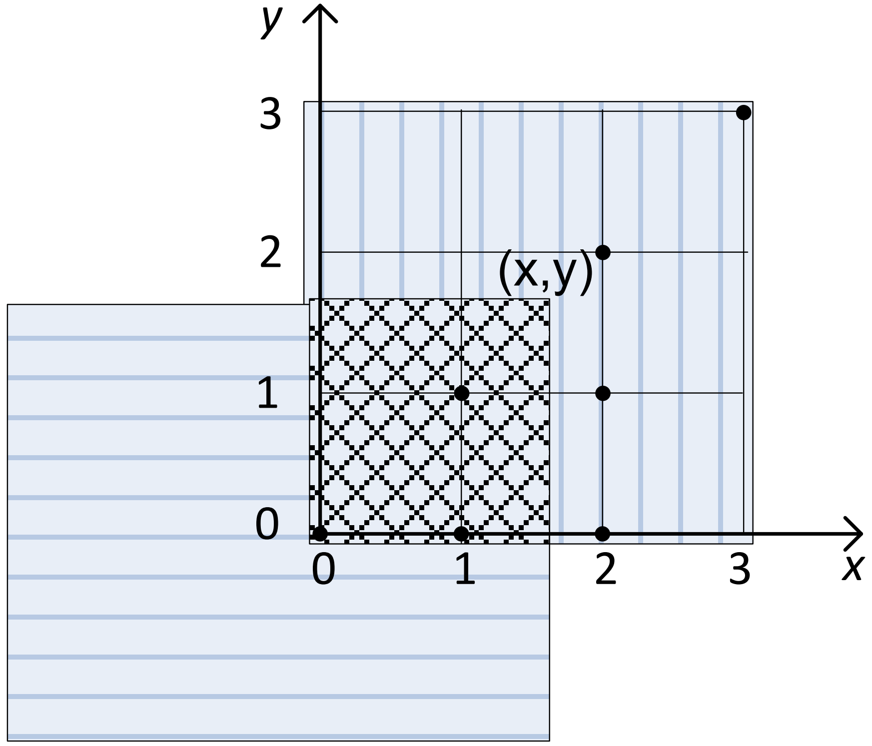
$P_{XY}(x,y)$	$y=0$	$y=1$	$y=2$	$y=3$	$P_X(x)$
$x=0$	1/64	0	0	0	1/64
$x=1$	6/64	3/64	0	0	9/64
$x=2$	9/64	9/64	9/64	0	27/64
$x=3$	0	0	0	27/64	27/64
$P_Y(y)$	16/64	12/64	9/64	27/64	1

Joint Cumulative Distribution Function

Definition 4.5 (JCDF)**.** The joint cumulative distribution function (JCDF) of random variables X and Y, denoted as $F_{XY}(x, y)$, is defined as

$$F_{XY}(x, y) = P[X \leq x \cap Y \leq y] = P[X \leq x, Y \leq y]$$

for all x and y. It is the probability of the intersection of two events: random variables X and Y therefore induce a probability distribution over a two-dimensional Euclidean plane.



y
3
2
1
0
(x,y)
0
1
2
3
x

$F_{XY}(x, y)$ has the following properties:

1. $F_{XY}(-\infty, -\infty) = F_{XY}(x, -\infty) = F_{XY}(-\infty, y) = 0$.
2. $F_{XY}(+\infty, +\infty) = 1$.
3. $$F_{XY}(x, y) = \sum_{i=-\infty}^{x} \sum_{j=-\infty}^{y} P_{XY}(i, j).$$

any x_1, x_2, y_1 and y_2 such that $x_1 < x_2$ and $y_1 < y_2$, the probability $P[x_1 < X \leq x_2, y_1 < Y \leq y_2]$ is given in terms of $F_{XY}(x, y)$ by

$$P[x_1 < X \leq x_2, y_1 < Y \leq y_2] = F_{XY}(x_2, y_2) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1).$$

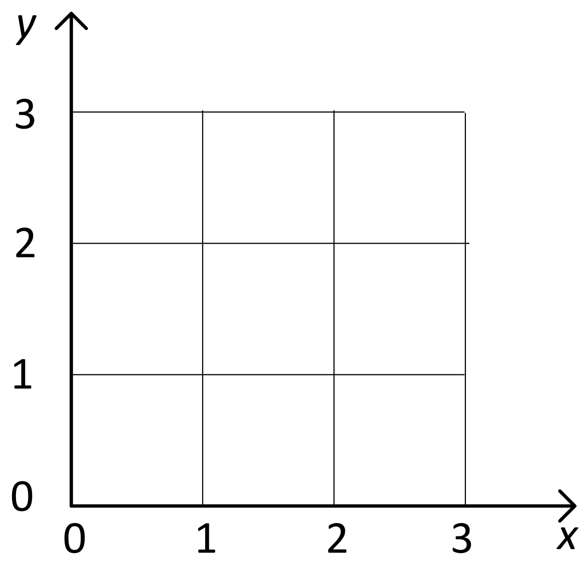
How to represent the JCDF

Single Random Variable:

The CDF $F_X(x)$ represents –

The sum of the PMFs of all points on the number line that are

On the line $X = x$, or to the left of the line $X = x$


y
3
2
1
0
0
1
2
3
x

Bivariate Random Variable:

The JCDF $F_{XY}(x, y)$ represents –

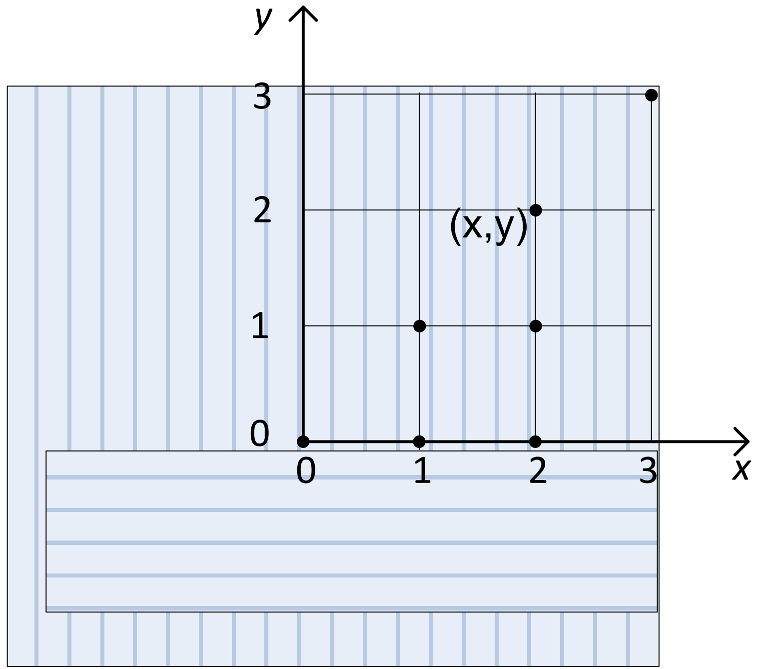
The sum of the JPMFs of all points on the 2-D plane that satisfy following two conditions:

1. On the line $X = x$, or to the left of the line $X = x$
2. On the line $Y = y$, or below the line $Y = y$

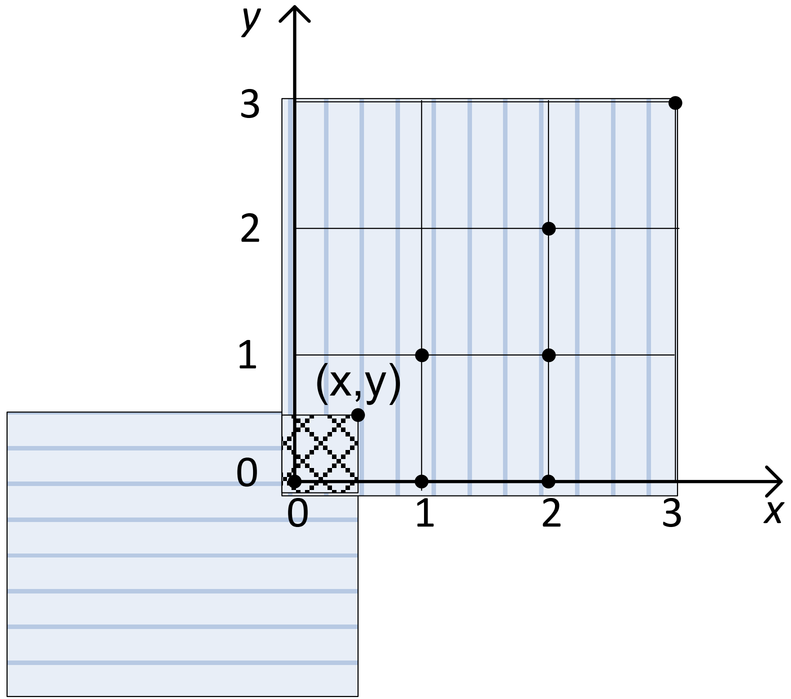
JCDF for random variable X and Y with JPDF given by

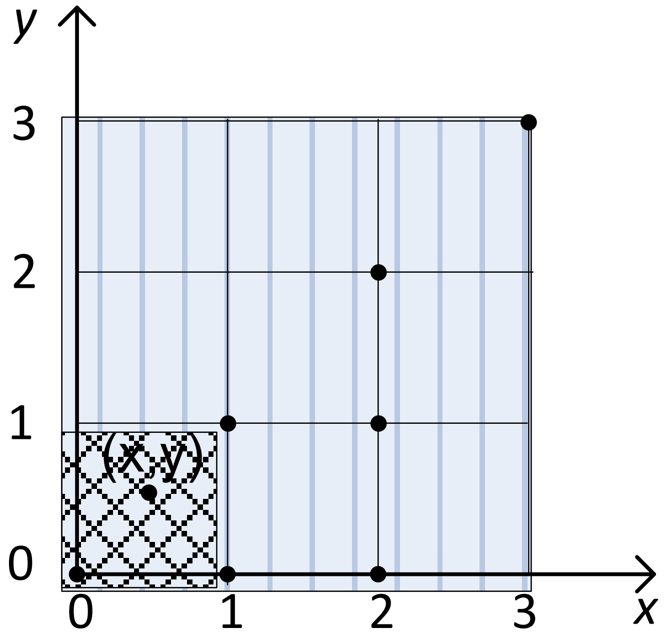
$$P_X(x) = \begin{cases} 27/64, & x = 3, y = 3 \\ 9/64\,, & x = 2, y = 2 \\ 9/64, & x = 2, y = 1 \\ 9/64\,, & x = 2, y = 0 \\ 3/64, & x = 1, y = 1 \\ 6/64, & x = 1, y = 0 \\ 1/64\,, & x = 0, y = 0 \\ 0, & \text{otherwise} \end{cases}$$

1. For $X < 0$ OR $Y < 0$

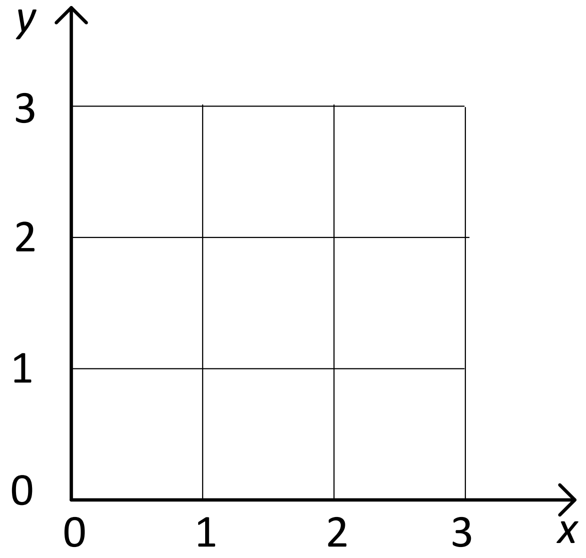


2. For $0 \leq X < 1$ AND $Y \geq 0$

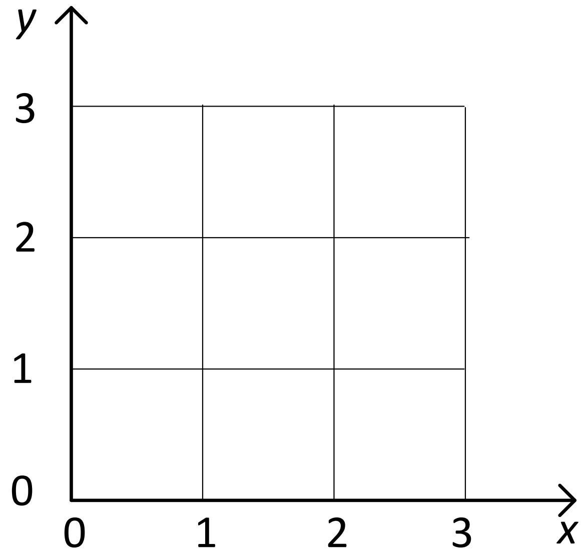




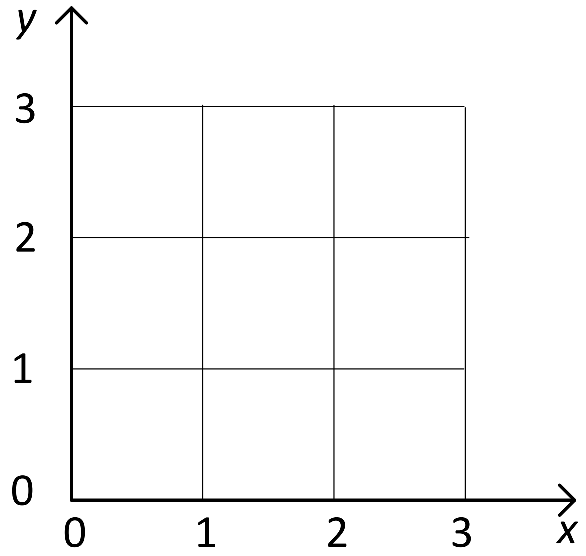
3. For $1 \leq X < 2$ AND $0 \leq Y < 1$



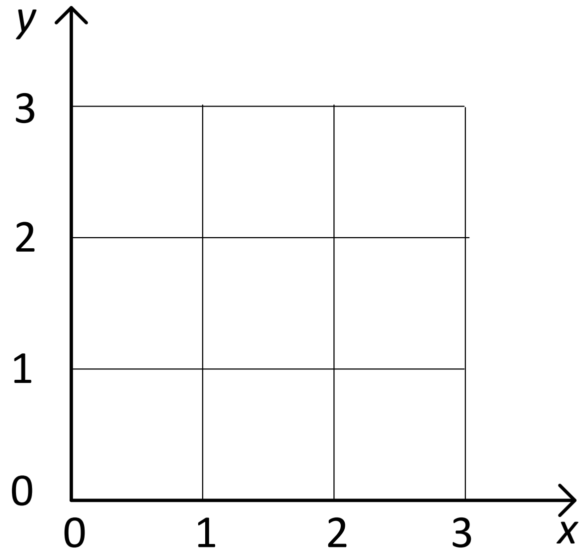
4. For $1 \leq X < 2$ AND $Y \geq 1$



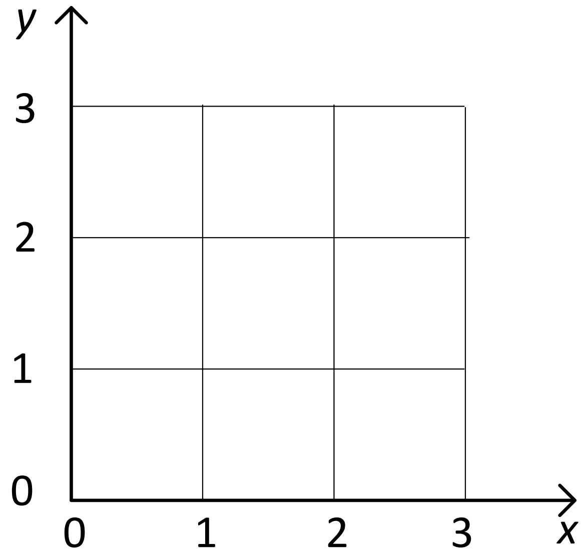
5. For $x \geq 2$ AND $0 \leq y < 1$



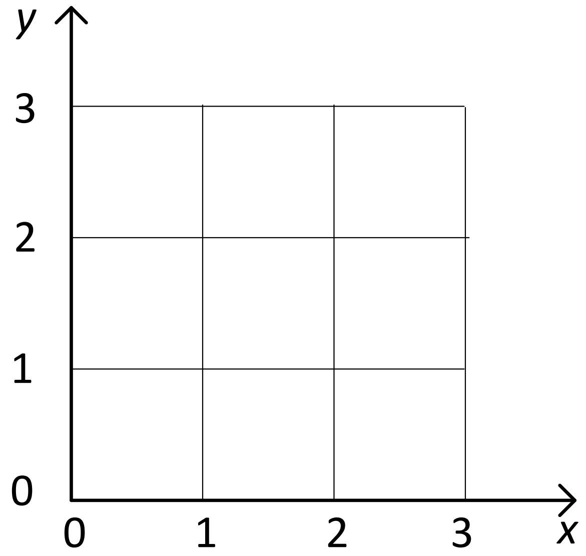
6. For $x \leq 2$ AND $1 \leq y < 2$



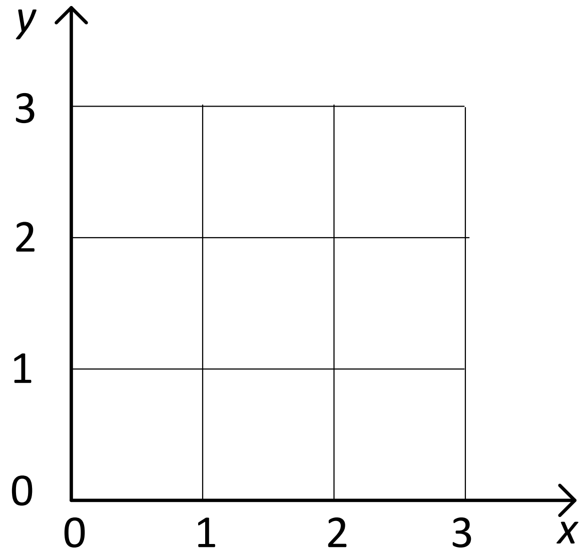
7. For $x \geq 2$ AND $2 \leq y < 3$



8. For $2 \leq x < 3$ AND $y \geq 3$



9. For $x \geq 3$ AND $y \geq 3$



Bivariate discrete probability models (JPMF and JCDF)

Example: Send 3 packets from one computer to another

Sample space $S = \{FFF, FFD, FDF, FDD, DFF, DFD, DDF, DDD\}$

Random variables:

X: # of successful deliveries $\quad S_X = \{0, 1, 2, 3\}$

Y: # of successful deliveries before any failure $\quad S_Y = \{0, 1, 2, 3\}$

Function that converts each $\omega \in S$ into a pair of real numbers

$g(DDD) = (3,3), \quad g(DDF) = (2,2), \quad g(DFD) = 2,1), \quad g(DFF) = (1,1)$

$g(FDD) = (2,0), \quad g(FDF) = (1,0), \quad g(FFD) = (1,0), g(FFF) = (0,0)$

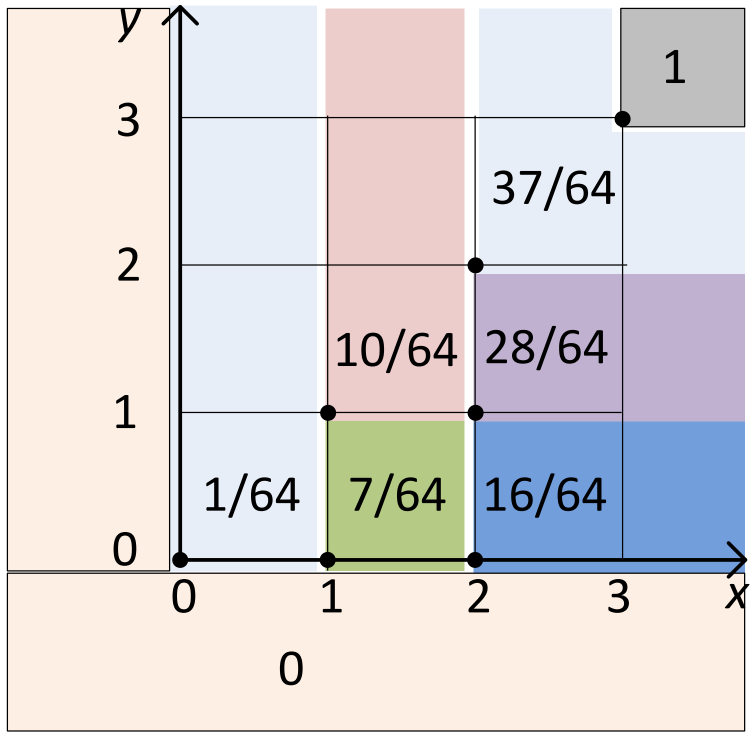
Probability associated with each pair of real numbers

$$P[DDD\} = \frac{27}{64}, P[DDF] = \frac{9}{64}, P[DFD] = \frac{9}{64}, P[DFF] = \frac{3}{64}, P[FDD] = \frac{9}{64},$$

$$P[FDF] = \frac{3}{64}, P[FFD] = \frac{3}{64}, P[FFF] = \frac{1}{64}$$

$$P_X(x) = \begin{cases} 27/64, & x = 3, y = 3 \\ 9/64\,, & x = 2, y = 2 \\ 9/64, & x = 2, y = 1 \\ 9/64\,, & x = 2, y = 0 \\ 3/64, & x = 1, y = 1 \\ 6/64, & x = 1, y = 0 \\ 1/64\,, & x = 0, y = 0 \\ 0, & \text{otherwise} \end{cases}$$

$P_{XY}(x,y)$	$y = 0$	$y = 1$	$y = 2$	$y = 3$	$P_X(x)$
$x = 0$	1/64	0	0	0	1/64
$x = 1$	6/64	3/64	0	0	9/64
$x = 2$	9/64	9/64	9/64	0	27/64
$x = 3$	0	0	0	27/64	27/64
$P_Y(y)$	16/64	12/64	9/64	27/64	1



$$F_X(x) = \begin{cases} 0, & x < 0 \text{ OR } y < 0 \\ \frac{1}{64}, & 0 \leq x < 1, y \geq 0 \\ \frac{7}{64}, & 1 \leq x < 2, 0 \leq y < 1 \\ \frac{10}{64}, & 1 \leq x < 2, y \geq 1 \\ \frac{16}{64}, & x \geq 2, 0 \leq y < 1 \\ \frac{28}{64}, & x \geq 2, 1 \leq y < 2 \\ \frac{37}{64}, & x \geq 2, 2 \leq y < 3 \\ & 2 \leq x < 3, y \geq 3 \\ 1, & x \geq 3, y \geq 3 \end{cases}$$

Properties of JPMF

1. $P_{XY}(x, y) \geq 0$
2. $\sum_x \sum_y P_{XY}(x, y) = 1$
3. $P_X(x) = \sum_y P_{XY}(x, y)$
4. $P_Y(y) = \sum_x P_{XY}(x, y)$

Properties of JCDF

$F_{XY}(x, y)$ has the following properties:

1. $F_{XY}(-\infty, -\infty) = F_{XY}(x, -\infty) = F_{XY}(-\infty, y) = 0.$

2. $F_{XY}(+\infty, +\infty) = 1.$

3. $$F_{XY}(x, y) = \sum_{i=-\infty}^{x} \sum_{j=-\infty}^{y} P_{XY}(i, j).$$

Marginal Cumulative Distribution Function

Definition 4.6. If X and Y are two discrete random variables with joint CDF $F_{XY}(x, y)$, then marginal CDF of X and Y are

$$\begin{aligned} F_X(x) &= P[X \leq x] \\ &= P[X \leq x \cap Y \leq +\infty] \\ &= F_{XY}(x, \infty). \end{aligned}$$

The above relation follows from the fact that the joint event $X \leq x \cap Y \leq \infty$ is the same as the event $X \leq x$, since $Y \leq \infty$ is a sure event. Similarly,

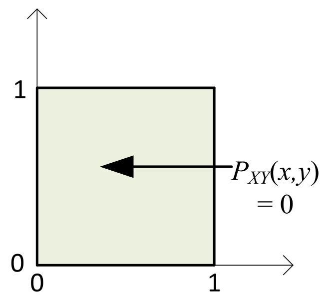
$$F_Y(y) = F_{XY}(\infty, y).$$

Bivariate Continuous Random Variables

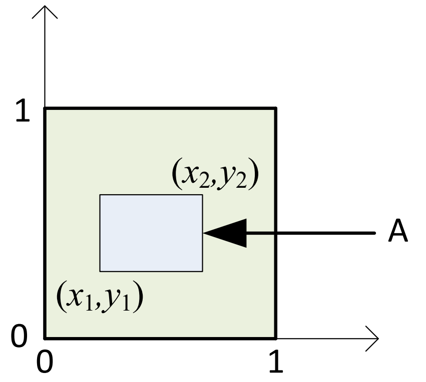
1. Like the PMF of a single continuous random variable is meaningless
    a. JPMF for joint continuous RVs is meaningless as well
2. JCDF for bivariate continuous RVs are well-defined
3. JPDF for bivariate continuous RVs are also well-defined
4. Marginal PDF and Marginal CDF are used as well

Joint Cumulative Distribution Function (JCDF)

Consider 2 continuous random variables X and Y jointly continuous within a unit square



$$P[X = x, Y = y] = \frac{1}{n}\bigg|_{n=\infty} = 0.$$



$$A = [x_1 \le X \le x_2, y_1 \le Y \le y_2 \in]$$

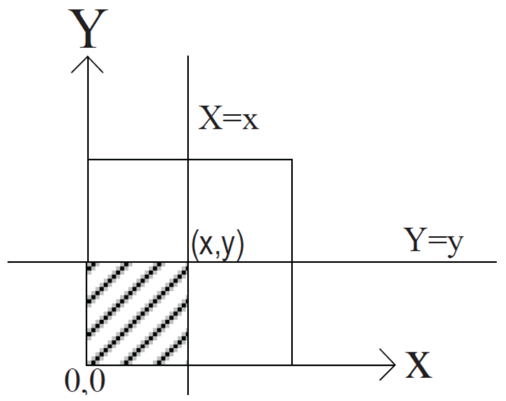
$$P[(x,y) \in A] > 0 \text{ if } 0 \le X \le 1, 0 \le Y \le 1$$

$$P[(x,y) \in A] = P[x_1 \le X \le x_2, y_1 \le Y \le y_2]$$

$$= \frac{\text{Area of the region } A}{\text{Total area of the square}} = (x_2 - x_1)(y_2 - y_1)$$

When $x_1 = -\infty, x_2 = x, y_1 = -\infty, y_2 = y$,

$$P[x_1 \le X \le x_2, y_1 \le Y \le y_2] = P[-\infty \le X \le x, -\infty \le Y \le y]$$
$$= P[X \le x, Y \le y] = F_{XY}(x, y)$$



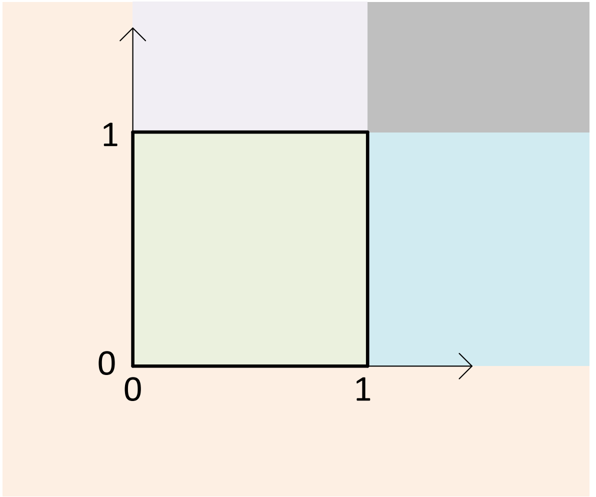
Definition 4.7. The joint cumulative distribution function (JCDF) of a pair of continuous random variables X and Y, defined on a single sample space, is denoted by $F_{XY}(x, y)$, and defined by

$$F_{XY}(x, y) = P[X \leq x, Y \leq y],$$

and it represents the joint probability of occurring two events $\{X \leq x\}$ and $\{Y \leq y\}$ simultaneously.

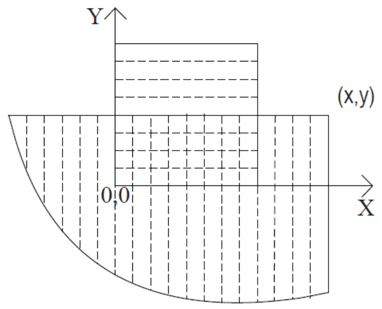
Mathematical Representation of the CDF

1. The joint CDF of a pair of continuous RVs has to be defined for all points on the $X - Y$ Plane.
2. The whole needs to be divided into sub-regions such that for each sub-region, a single function can define the CDF



How to calculate the JCDF for a pair of continuous RVs

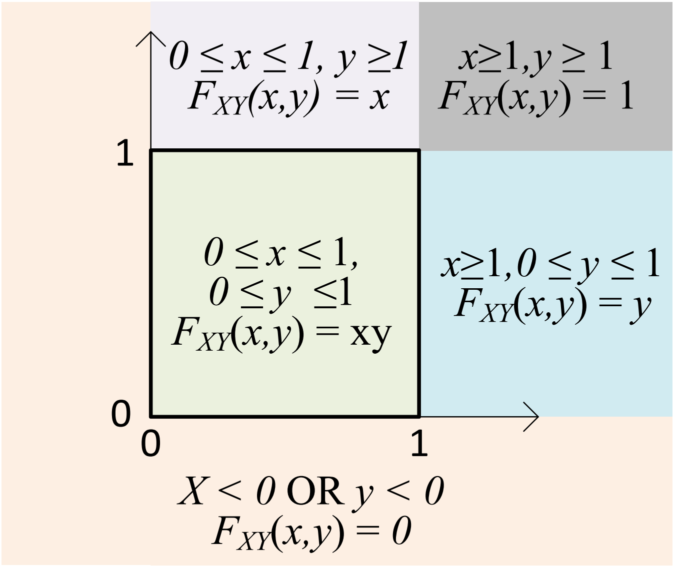
1. Let X and Y are uniformly distributed within a unit square
2. CDF of the point (x, y), $F_{XY}(x, y)$ is to be calculated
3. Let S_{XY} denote the area for which X and Y are defined
   $$S_{XY} = [0 \le X \le 1, 0 \le Y \le 1]$$
4. JCDF for (x, y) defines the probability for a region, say R,
   $$R = [-\infty < X \le x, -\infty < Y \le 1]$$



5. JCDF defines the probability for the intersection of two regions, $S_{XY} \cap R$

$$S_{XY}(x, y) = S_{XY} \cap R$$

$$F_{XY}(x, y) = P[X \leq x, Y \leq y] = P[(x, y) \in S_{XY}(x, y)$$



1. Whenever either of the random variables is less than zero, the intersection of the area by the regions $[-\infty < X \le x, -\infty < Y \le y]$ and $[0 \le X \le 1, 0 \le Y \le 1]$ is empty. Therefore, if $X < 0$ or $Y < 0$ then the JCDF is zero, i.e.,

$$F_{X,Y}(x,y) = 0, \quad \text{if} \quad x < 0 \quad \text{or} \quad y < 0.$$

The JCDF of any point where $x > 1$ and $0 \le y < 1$ will include the double shared region with area $r_2 = y$ and the JCDF is given by

$$F_{X,Y}(x,y) = y \quad \text{if} \quad x > 1 \quad \text{or} \quad 0 \le y < 1.$$

Any point within the shaded region will have a JCDF given by

$$F_{X,Y}(x,y) = x \quad \text{if} \quad y > 1 \quad \text{or} \quad 0 \le x < 1.$$

The JCDF of any point (x,y) within the unit square can be represented by a single function The JCDF is the area of the intersection of the unit square $[0 \le x \le 1, 0 \le y \le 1]$ and area covered by $F_{X,Y}(x,y)$ which is $[-\infty < X \le x, -\infty \le Y \le y]$, which is $r_3 = x_4$ and the JCDF is given by

$$F_{X,Y}(x,y) = xy \quad \text{if} \quad 0 \le x \le 1, \quad 0 \le y \le 1.$$

The JCDF of any point within the shaded region can be represented by a single function $r_5 = 1$ and the JCDF is given by

$$F_{X,Y}(x,y) = 1 \quad \text{if} \quad x > 1, \quad y > 1.$$

Finally, the complete JCDF is given by

$$F_X, Y(x, y) = \begin{cases} 0, & \text{if } x < 0 \text{ or } y < 0; \\ y, & \text{if } x > 1, 0 \leq y \leq 1; \\ x, & \text{if } 0 \leq x \leq 1, y > 1; \\ xy, & \text{if } 0 \leq x \leq 1, 0 \leq y \leq 1; \\ 1, & \text{if } x > 1, y > 1 \end{cases}$$

Properties of JCDF

i) The joint cumulative distribution function is a measure of probability, it must take on a value between 0 and 1, i.e.,

$$0 \leq F_{X,Y}(x, y) \leq 1$$

ii) The random variables X and Y must assume real value. It is impossible for either of these two to take a value less than or equal to $-\infty$, and both must be less than $+\infty$. Therefore,

$$F_{X,Y}(-\infty, y) = 0$$

$$F_{X,Y}(x, -\infty) = 0$$

$$F_{X,Y}(-\infty, -\infty) = 0$$

$$F_{X,Y}(+\infty, +\infty) = 1$$

iii) The probability that X and Y have values within certain intervals can be calculated from their JCDF as shown in Fig. -.- and is given by

$$F_{X,Y}(x_1 \leq X < x_2, y_1 \leq Y < y_2) = F_{X,Y}(x_2, y_2) - F_{X,Y}(x_1, y_2) - F_{X,Y}(x_2, y_1) + F_{X,Y}(x_1, y_1)$$

Joint Probability Density Function

The joint probability density function (JPDF) of jointly distributed random variables X and Y at point (x, y) is denoted by $f_{XY}(x, y)$ and is given by

$f_{XY}(x, y) =$

$$f_{X,Y}(x, y) = \begin{cases} 1, & \text{for } 0 < x < 1, 0 < y < 1; \\ 0, & \text{otherwise.} \end{cases}$$

Properties of JPDF

i. Since the JPDF represents a probability (i.e., probability per unit area), it must be nonnegative.

$$f_{X,Y}(x,y) \geq 0.$$

ii. Since the JPDF represents probability per unit area, the JPDF multiplied by by the total area will give the total probability which is 1.

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f_{X,Y}(x,y)dydx = 1.$$

iii. An extension of the above property

$$F_{X,Y}(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(x,y)dydx$$

iv. The probability of getting the pair of values in any bounded region is

$$P[x_1 < X \leq x_2, y_1 < Y \leq y_2] = \int_{x_1}^{x_2} \int_{y_1}^{y_2} f_{X,Y}(x,y)dydx$$

v. If the two random variables are jointly uniformly distributed over the whole area associated with the random variables, then the JPDF is a constant.

$f_{X,Y}(x,y) = c.$

Example 4.6. Random variable X and Y have the joint PDF given by

$$f_{X,Y}(x,y) = \begin{cases} c, & \text{for } 0 \leq x \leq 5, 0 \leq y \leq 3; \\ 0, & \text{otherwise.} \end{cases}$$

a) Find the value of the constant c.

b) Find the probability $P[A] = P[2 \leq X \leq 3, 1 \leq Y \leq 3]$.

c) Find the probability $P[B] = P[Y \geq X]$.

$$\int_0^5 \int_0^3 f_{X,Y}(x,y)dydx = 1$$
$$\Rightarrow \int_0^5 \int_0^3 cdydx = 1$$
$$\Rightarrow \int_0^5 \left[cy\Big|_0^3 \right] dx = 1$$
$$\Rightarrow \int_0^5 \quad 3cdx = 1$$
$$\Rightarrow 3cx\Big|_0^5 = 3c(5-0) = 1$$
$$\Rightarrow 15c = 1$$
$$\Rightarrow c = \frac{1}{15}.$$

$$
\begin{aligned}
P[A] &= \int_2^3 \int_1^3 \frac{1}{15} dy dx \\
&= \int_2^3 \left[\frac{y}{15} \Big|_1^3 \right] dx \\
&= \int_2^3 \frac{3-1}{15} dx \\
&= \frac{2x}{15} \Big|_2^3 \\
&= \frac{2}{15}(3-2) \\
&= \frac{2}{15}.
\end{aligned}
$$

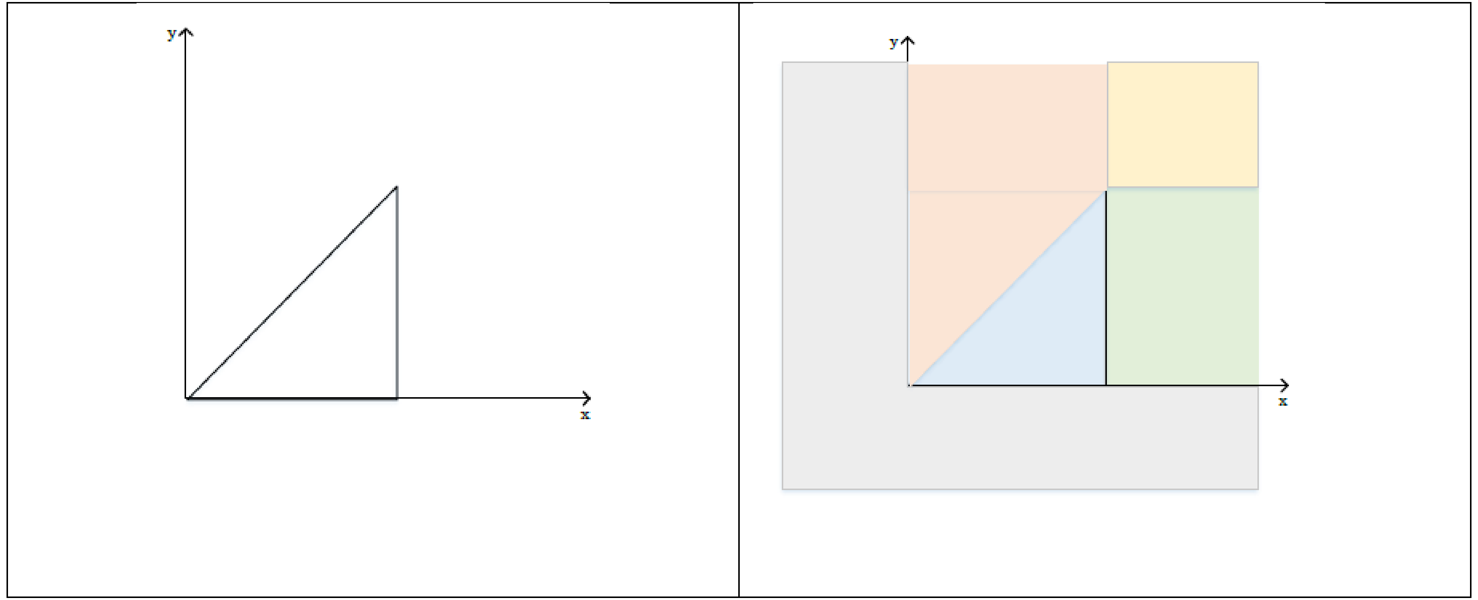
Oppositely, the same probability can be found by calculating the ratio of the desired area to the total area. The area of the desired region is $A = 1 \times 2 = 2$ and the total area is $5 \times 3 = 15$. Therefore, the probability is

$$P[A] = \frac{\text{Area of } A}{\text{Total Area}} = \frac{2}{3}.$$

$$
\begin{aligned}
P[B] &= \int_0^3 \int_x^3 \frac{1}{15} dy dx \\
&= \int_0^3 \frac{y}{15}\Big|_x^3 dx \\
&= \int_0^3 \frac{3-x}{15} dx \\
&= \frac{3}{15}x\Big|_0^3 - \frac{1}{30}x^2\Big|_0^3 \\
&= \frac{9}{15} - \frac{9}{30} \\
&= \frac{9}{15} = \frac{3}{5}.
\end{aligned}
$$

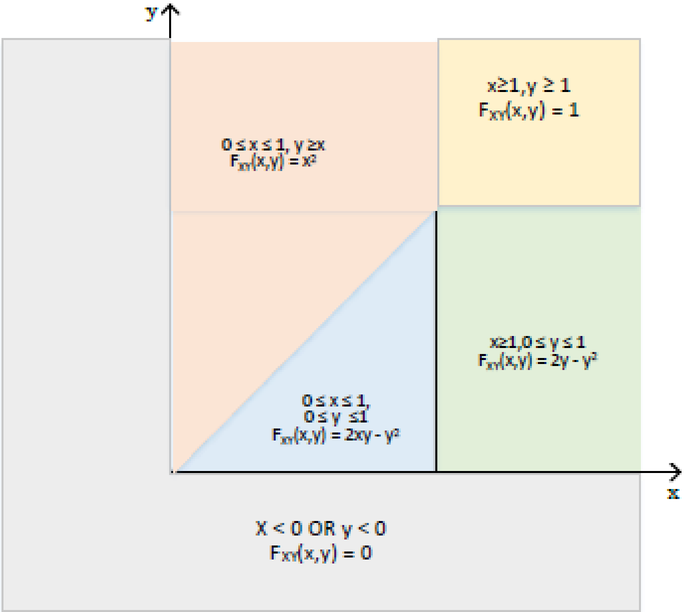
Example 7. Continuous random variables X and Y are defined to be jointly uniform within the region $0 \leq y \leq x \leq 1$.

a) Find the joint PDF of X and Y, $f_{XY}(x, y)$.
b) Find the joint CDF of X and Y, $F_{XY}(x, y)$.

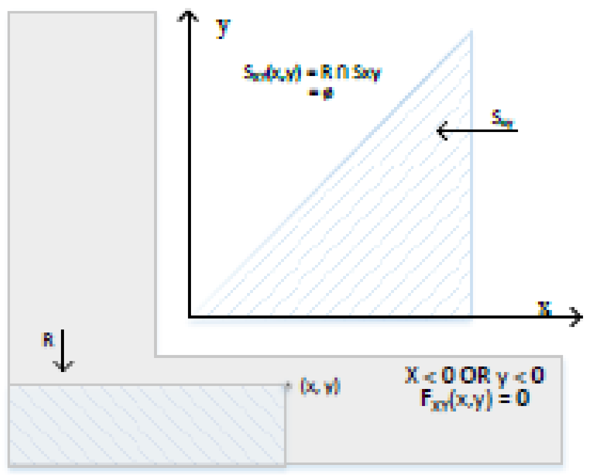


The joint pdf of X and Y $f_{XY}(x, y)$ is given by

$$f_{XY}(x, y) = \begin{cases} 2, & 0 \le y \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

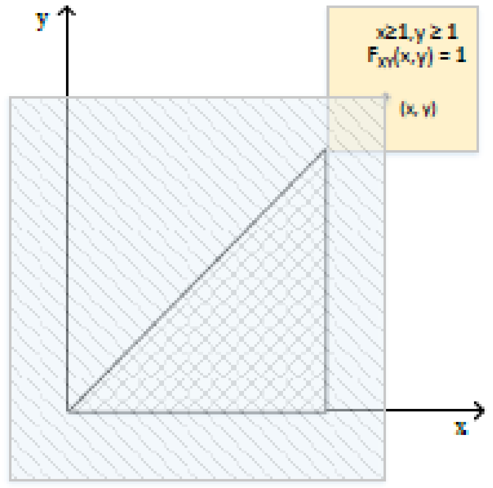


Case 1: $x < 0$ OR $y < 0$



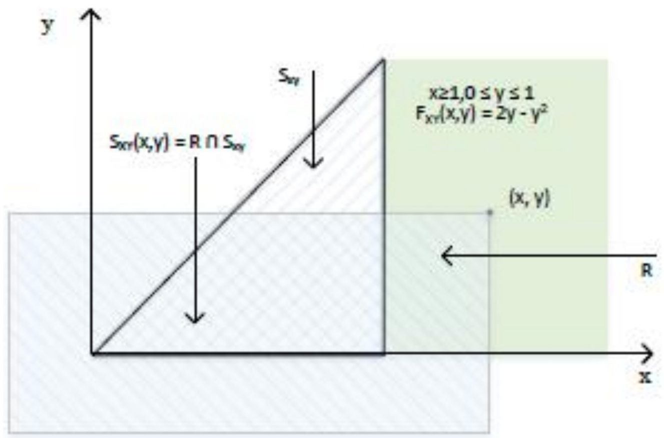
$F_{XY}(x, y) = 0$

Case 5: $x \geq 1, y \geq 1$



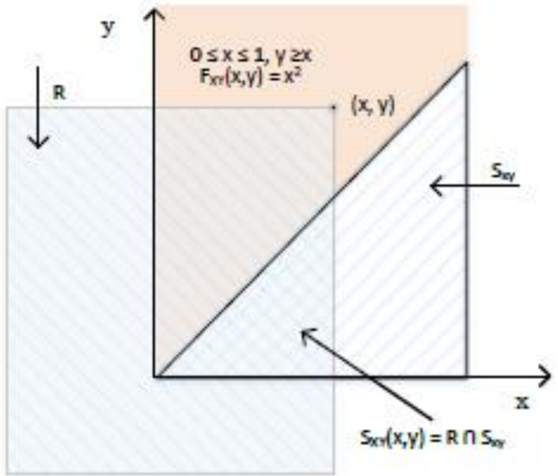
$F_{XY}(x, y) = 1$

Case 2: $x \geq 1, 0 \leq y < 1$



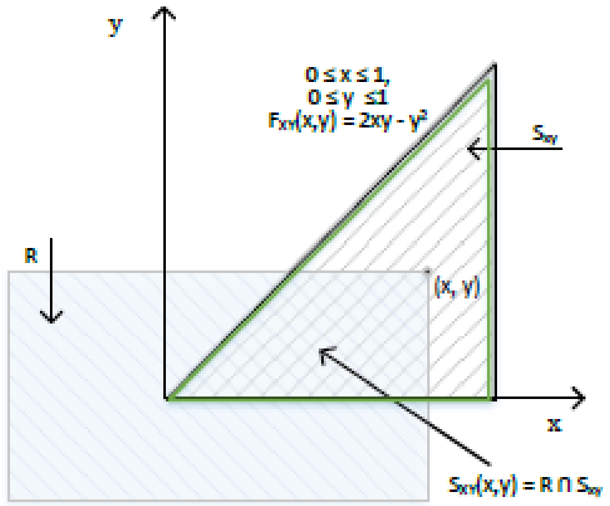
$$
\begin{aligned}
F_{X,Y}(x, y) &= \int_0^y \int_y^1 2dxdy \\
&= \int_0^y \left[2x \Big|_y^1 \right] dy \\
&= \int_0^y 2(1 - y)dy \\
&= \int_0^y 2dy - \int_0^y 2ydy \\
&= 2y \Big|_0^y - \frac{2y^2}{2} \Big|_0^y \\
&= 2y - y^2.
\end{aligned}
$$

Case 3: $0 \leq x < 1, y \geq x$



$$
\begin{aligned}
F_{X,Y}(x,y) &= \int_0^x \int_y^x 2dxdy \\
&= \int_0^x \left[2x \Big|_y^x \right] dy \\
&= \int_0^x (2x - 2y)dy \\
&= 2xy\Big|_0^x - \frac{2y^2}{2}\Big|_0^x \\
&= 2x^2 - x^2 \\
&= x^2.
\end{aligned}
$$

Case 4: $0 \leq x < 1, 0 \leq y < 1$



$$
\begin{aligned}
F_{X,Y}(x,y) &= \int_0^y \int_y^x 2dxdy \\
&= \int_0^y \left[2x \Big|_y^x \right] dy \\
&= \int_0^y (2x - 2y)dy \\
&= 2xy\Big|_0^y - \frac{2y^2}{2}\Big|_0^y \\
&= 2xy - y^2.
\end{aligned}
$$

The complete JCDF of the two RVs is

$$F_{X,Y}(x,y) = \begin{cases} 0, & \text{for } x < 0 \text{ or } y < 0; \\ 2y - y^2, & \text{for } x > 1, 0 \le y \le 1; \\ x^2, & \text{for } 0 \le x \le 1, y \ge x; \\ 2xy - y^2, & \text{for } 0 \le y \le x \le 1; \\ 1, & \text{for } x > 1, y > 1. \end{cases}$$

Marginal probability density function

Definition 4.9. *Marginal Probability Density Function.* If X and Y are random variable with joint PDF $f_{X,Y}(x, y)$, then the marginal PDF of X and Y are given by

$$f_X(x) = \int_{-\infty}^{\infty} f_{x,y}(x, y)dy, \tag{4.28}$$

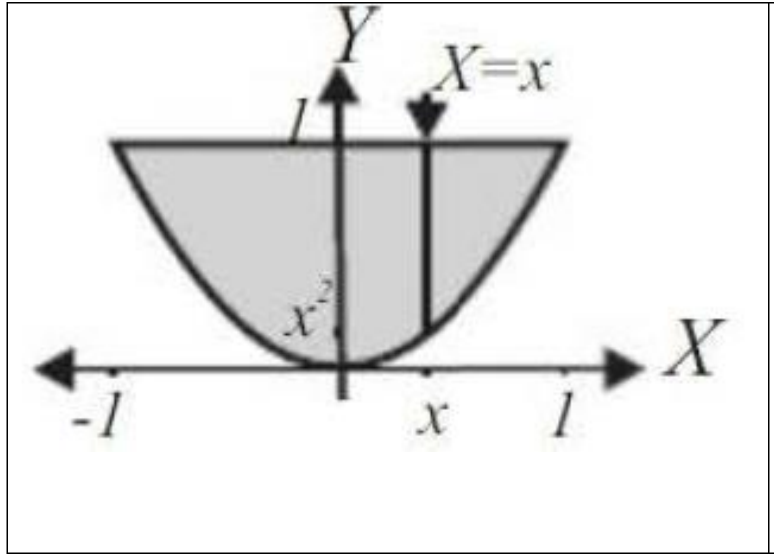
and

$$f_Y(y) = \int_{-\infty}^{\infty} f_{x,y}(x, y)dx. \tag{4.29}$$

Example 4.8. The joint PDF of X and Y is given by

$$f_{X,Y}(x,y) = \begin{cases} \frac{5}{4}y, & \text{for } -1 \le x \le 1, x^2 \le y \le 1; \\ 0, & \text{otherwise.} \end{cases}$$

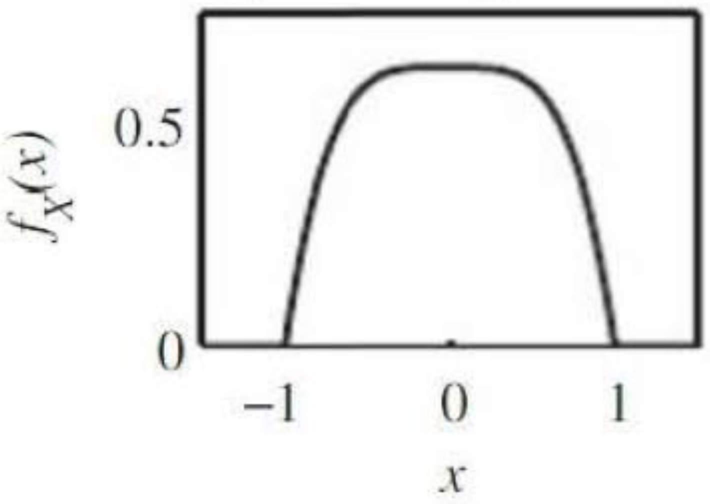
Find the marginal PDFs of X and Y.

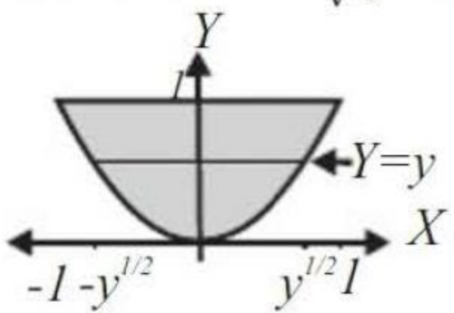


$$
\begin{aligned}
f_X(x) &= \int_{x^2}^{1} f_{X,Y}(x,y)dy \\
&= \int_{x^2}^{1} \frac{5}{4} y dy \\
&= \frac{5}{8} y^2 \Big|_{x^2}^{1} \\
&= \frac{5}{8}(1 - x^4).
\end{aligned}
$$

The complete expression for marginal PDF of X is

$$
f_X(x) = \begin{cases} \frac{5}{8}(1 - x^4), & \text{for } -1 \leq x \leq 1; \\ 0, & \text{otherwise.} \end{cases}
$$

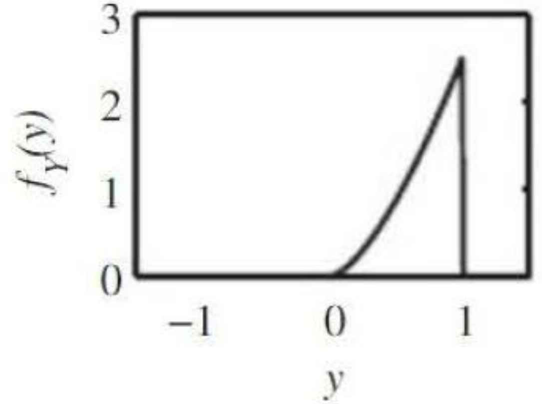




$$
\begin{aligned}
f_Y(y) &= \int_{-\sqrt{y}}^{+\sqrt{y}} \frac{5}{4} y\,dx \\
&= \frac{5}{4} yx \Big|_{-\sqrt{y}}^{+\sqrt{y}} \\
&= \frac{5}{4} y(\sqrt{y} + \sqrt{y}) \\
&= \frac{5}{2} \cdot y \cdot y^{\frac{1}{2}} \\
&= \frac{5}{2} \cdot y^{\frac{3}{2}}.
\end{aligned}
$$

The complete expression of the marginal PDF of Y is given by

$$
f_Y(y) = \begin{cases} \frac{5}{2} y^{\frac{3}{2}}, & \text{for } 0 \le y \le 1; \\ 0, & \text{otherwise.} \end{cases}
$$



Example 4.9. The joint PDF of random variables X and Y is given by

$$f_{X,Y}(x,y) = \begin{cases} \lambda x y^2, & \text{for } 0 \le x \le y \le 1; \\ 0, & \text{otherwise.} \end{cases}$$

1. Determine the value of λ.

2. Find the marginal probability density function of X and Y.

3. Find $E[X]$ and $E[Y]$.

Value of λ

To find λ, note that $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$ gives

$$\int_0^1 \left(\int_x^1 \lambda x y^2\,dy\right) dx = 1.$$

Therefore,

$$\begin{aligned}
1 &= \int_0^1 \left(\int_x^1 y^2\,dy\right)\lambda x\,dx = \int_0^1 \left[\frac{1}{3}y^3\right]_x^1 \lambda x\,dx \\
&= \lambda \int_0^1 \left(\frac{1}{3} - \frac{1}{3}x^3\right) x\,dx = \frac{\lambda}{3}\int_0^1 (1-x^3)x\,dx = \frac{\lambda}{3}\int_0^1 (x-x^4)\,dx \\
&= \frac{\lambda}{3}\left[\frac{1}{2}x^2 - \frac{1}{5}x^5\right]_0^1 = \frac{\lambda}{10},
\end{aligned}$$

and hence $\lambda = 10$.

Marginal PDF of X

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy = \int_x^1 10xy^2\,dy = \left[\frac{10}{3}xy^3\right]_x^1 = \frac{10}{3}x(1-x^3),\ \ 0 \le x \le 1$$

Marginal PDF of Y

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx = \int_0^y 10xy^2\,dx = \left[5x^2y^2\right]_0^y = 5y^4, \quad 0 \le y \le 1$$

Multiple Random Variables: Multivariate Joint Distribution

- Random Experiments with more than two RVs
- Outcomes of the experiments are represented by points on an n dimensional space

The probability models associated are

1. Multivariate joint PMF (MJPMF) – represents the probability of occurring an outcome $x_1, \dots, x_n$ – simultaneously n RVs have the values (as real numbers)
   $X_1 = x_1, \dots, X_n = x_n$
2. Multivariate Joint PDF (MJPDF) – represents the probability density at a point $(x_1, \dots, x_n)$ in the n-dimensional space
3. Multivariate Joint CDF (MJCDF) – represents the total probability of an n-dimensional space associated with the point $(x_1, \dots, x_n)$ and defined by the lines $X_1 = x_1, \dots, X_n = x_n$.

Marginal PMFs/JPMFs/MJPMFs, Marginal PDFs/JPDFs/MJPDFs and marginal CDFs/JCDFs/MJCDFs of all the subsets of RVs

Multiple Discrete RVs: Multivariate Joint Discrete Distributions

Multivariate Joint Probability Mass Function (MJPMF)

$$P_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = P[X_1 = x_1,\ldots,X_n = x_n].$$

Properties:

1. $P_{X_1,\ldots,X_n}(x_1,\ldots,x_n) \geq 0.$
2. If for some i, $1 \leq i \leq n$, $x_i \notin S_{X_i}$, then
$$P_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = 0.$$
3. $\sum_{x_i \in S_{X_i}, 1 \leq i \leq n} P_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = 1.$

Multivariate Joint CDF

$$
\begin{aligned}
F_{X_1,\ldots,X_n}(x_1,\ldots,x_n) &= P[X_1 \leq x_1,\ldots,X_n \leq x_n], \quad x_i < +\infty, 1 \leq i \leq n \\
&= \sum_{x_1 \in S_{X+1}} \sum_{x_j \in S_{X_j}}^{x_i} P_{X_1,\ldots,X_n}(x_1,\ldots,x_n)
\end{aligned}
$$

Lower part need to be corrected

Marginal PMF for multivariate distributions

1. Marginal PMF
2. Marginal Joint PMF
3. Marginal Multivariate Joint PMF

Marginal PMF of X_i

$$
\begin{aligned}
P_{X_i}(x_i) &= P[X_i = x_i] \\
&= P[X_i = x_i, x_j \in S_{X_j}], \quad i \le j \le n, i \ne j \\
&= \sum_{x_1 \in S_{X_1}} \cdots \sum_{x_{i-1} \in S_{X_{i-1}}} \sum_{x_{i+1} \in S_{X_{i+1}}} \cdots \sum_{x_n \in S_{X_n}} P_{X_1,\dots,X_n}(x_1, \dots, x_n), \qquad 1 \le i \le n
\end{aligned}
$$

Marginal Joint PMF of X_j and X_k

Let I is the set of indices of the RVs, $I = \{1, 2, \dots, n\}$

The marginal Joint PMF of any pair of RVS with indices $J = \{j, k\}, J \subset I$

$$P_{X_j X_k}(x_j, x_k) = P[X_j = x_j, X_k = x_k]$$

$$= \sum_{x_{i_1} \in S_{X_{i_1}}} \cdots \sum_{x_{i_{n-2}} \in S_{X_{i_{n-2}}}} P_{X_1, \dots, X_n}(x_1, \dots, x_n), \qquad i_1, \dots, i_{n-2} \in I, i_1, \dots, i_{n-2} \notin J$$

The Marginal Multivariate Joint PMF of any subset of RVS from I with indices $J = \{k_1, \dots, k_m\}, J \subset I$ and $m < n$

$$P_{X_j X_k}(x_{k_1}, \dots, x_{k_m}) = P[X_{k_1} = x_{k_1}, \dots, X_{k_m} = x_{k_m}]$$

$$= \sum_{x_{i_1} \in S_{X_{i_1}}} \cdots \sum_{x_{i_{n-2}} \in S_{X_{i_{n-m}}}} P_{X_1, \dots, X_n}(x_1, \dots, x_n), \quad i_1, \dots, i_{n-m} \in I, i_1, \dots, i_{n-m} \notin J$$

Marginal CDF multiple RVs

$$\begin{aligned}
F_{X_i}(x) &= P[X_i \leq x] \\
&= P[X_1 < \infty, \ldots, X_{i-1} < \infty, X_i = x, X_{i+1} < \infty, X_n < \infty] \\
&= \sum_{x_1 < \infty} \cdots \sum_{x_{i-1} < \infty} \sum_{x_i < x} \sum_{x_{i+1} < \infty} \cdots \sum_{x_n < \infty} P_{X_1, \ldots, X_n}(x_1, \ldots, x_n) \\
&= \lim_{\substack{x_j \to \infty \\ 1 \leq j \leq n, j \neq i}} F_{X_1, \ldots, X_n}(x_1, \ldots, x_n).
\end{aligned}$$

Example: In a box, there are 23 balls of which 5 balls are red, 8 balls are blue and 10 balls are green. Suppose 7 balls are picked from the box without replacement. Let X, Y and Z represent the number of red, blue and green balls picked. Find the multivariate joint PMF and marginal PMFs of X, Y and Z.

Let the multivariate joint PMF is denoted by $P_{XYZ}(x, y, z)$

$$P_{XYZ}(x, y, z) = \frac{\#\ of\ ways\ exactly\ x\ red, y\ blue\ and\ z\ green\ balls\ picked}{\#\ of\ ways\ 7\ balls\ are\ picked\ from\ 23\ balls}$$

$$= \frac{a}{b}$$

The probability represented by $P_{XYZ}(x, y, z)$ is the ratio of a to b, where:

1. a counts the number of ways x red, y blue and z green balls can be picked which is the product of three terms $p, q,$ and r, $a = pqr$, where
    a. p counts the number of ways x red balls are picked from 5 red balls, $p = \binom{5}{x}$
    b. q counts the number of ways y blue balls are picked from 8 blue balls, $q = \binom{8}{y}$
    c. r counts the number of ways z green balls are picked from 10 green balls, $r = \binom{10}{z}$

$$a = pqr = \binom{5}{x}\binom{8}{y}\binom{10}{z}$$

2. b counts the number of ways 7 balls are picked from 23 balls, $b = \binom{23}{7}$

Therefore, the MJPMF of X, Y and Z $P_{X,Y,Z}(x, y, z)$, for $0 \leq x \leq 5, 0 \leq y \leq 8, 0 \leq z \leq 10$, and $x + y + z = 7$, can be given by

$$P_{X,Y,Z}(x, y, z) = \frac{\binom{5}{x}\binom{8}{y}\binom{10}{z}}{\binom{23}{7}}$$

and $P_{X,Y,Z}(x, y, z) = 0$ otherwise.

$$P_{XYZ}(x, y, z) = \begin{cases} \dfrac{\binom{5}{x}\binom{8}{y}\binom{10}{z}}{\binom{23}{7}}, & 0 \leq x \leq 5 \\ & 0 \leq y \leq 8 \\ & 0 \leq z \leq 10 \\ & x + y + z = 7 \\ 0, & \text{otherwise} \end{cases}$$

Marginal PMFs of X, Y and Z

The marginal PMF of X, $P_X(x)$, can be obtained in the following way:

$$\begin{aligned}
P_X(x) &= P[X = x] \\
&= \sum_{\substack{x+y+z=7 \\ 0\le y\le 7 \\ 0\le z\le 7}} P_{X,Y,Z}(x, y, z) \\
&= \sum_{y=0}^{7-x} \frac{\binom{5}{x}\binom{8}{y}\binom{10}{7-x-y}}{\binom{23}{7}} \\
&= \frac{\binom{5}{x}}{\binom{23}{7}} \sum_{y=0}^{7-x} \binom{8}{y}\binom{10}{7-x-y}.
\end{aligned}$$

$\sum_{y=0}^{7-x} \binom{8}{y}\binom{10}{7-x-y}$ is the total number of ways $(7-x)$ blue or green balls can be picked from 18 balls

Which is, number of ways $(7-x)$ balls are picked from 18 balls and is $\binom{18}{7-x}$

Therefore, the marginal PMF of X is

$$P_X(x) = \frac{\binom{5}{x}\binom{18}{7-x}}{\binom{23}{7}}.$$

Similarly,

$$P_Y(y) = \frac{\binom{8}{y}\binom{18}{7-y}}{\binom{23}{7}}$$

$$P_Z(z) = \frac{\binom{10}{z}\binom{18}{7-z}}{\binom{23}{7}}.$$

Multivariate Continuous Random Variables

Multivariate Joint CDF (MJCDF)

$$F_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = P[X_1 \le x_1,\ldots,X_n \le x_n]$$

If the joint distribution of n (where $n > 2$) random variables $X_1,\ldots,X_n$ is continuous, then the multivariate joint PDF, $f_{X_1,\ldots,X_n}(x_1,\ldots,x_n)$, can be derived from the MJCDF using the following relation

$$f_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = \frac{\partial^n F_{X_1,\ldots,X_n}(x_1,\ldots,x_n)}{\partial x_1,\ldots,\partial x_n},$$

$$F_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = \int_{-\infty}^{x_1}\cdots\int_{-\infty}^{x_n} f_{X_1,\ldots,X_n}(x_1,\ldots,x_n)dx_1\ldots dx_n.$$

Marginal density function for multiple RVs

If the multivariate joint PDF of $X_1, \ldots, X_n$ is $f_{X_1,\ldots,X_n}(x_1, \ldots, x_n)$, then the marginal PDF of X_1, $f_{X_1}(x_1)$, can be given by

$$f_{X_1}(x_1) = \underbrace{\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty}}_{n-1} f_{X_1,\ldots,X_n}(x_1, \ldots, x_n) dx_2 \ldots dx_n.$$

If the multivariate Joint PDF of 4 random variables $X_1, X_2, \ldots, X_4$ is $f_{X_1,\ldots,X_4}(x_1, \ldots, x_4)$, then

The marginal PDF of X_1, $f_{X_1}(x_1)$, is given by

$$f_{X_1}(x_1) = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} f_{X_1,\dots,X_4}(x_1,\dots,x_4)\ dx_2 dx_3 dx_4$$

The marginal joint PDF of X_1 and X_2, $f_{X_1X_2}(x_1,x_2)$, is given by

$$f_{X_1X_2}(x_1,x_2) = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} f_{X_1,\dots,X_4}(x_1,\dots,x_4)\ dx_3 dx_4$$

The marginal multivariate joint PDF X_1, X_2 and X_3, $f_{X_1,\dots,X_3}(x_1,\dots,x_3)$ is given by

$$f_{X_1,\dots,X_3}(x_1,\dots,x_3) = \int_{-\infty}^{+\infty} f_{X_1,\dots,X_4}(x_1,\dots,x_4)\, dx_4$$

Example 5.2. Suppose that a machine has three parts, and part i will fail at time X_i for $i = 1, 2, 3$. The following function may be the multivariate joint CDF of X_1, X_2 and X_3.

$$F_{X_1,X_2,X_3}(x_1, x_2, x_3) = \begin{cases} (1 - e^{-x_1})(1 - e^{-2x_2})(1 - e^{-3x_3}), & \text{for } x_1, x_2, x_3 \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

a) Find the multivariate joint PDF of X_1, X_2, X_3.

b) Find the marginal joint PDF of X_1 and X_2.

a) The multivariate joint PDF of X_1, X_2, X_3 is given by

$$\begin{aligned} f_{X_1,X_2,X_3}(x_1,x_2,x_3) &= \frac{\partial^3 F_{X_1,X_2,X_3}(x_1,x_2,x_3)}{\partial x_1 \partial x_2 \partial x_2} \\ &= e^{x_1}.2e^{2x_2}.3e^{3x_3} \\ &= 6e^{-x_1-2x_2-3x_3}, \quad x_1,x_2,x_3 \geq 0. \end{aligned}$$

Failure Times. We can find the marginal bivariate c.d.f. of X_1 and X_3 from the joint c.d.f. in Example 3.7.2 by letting x_2 go to ∞. The limit is

$$F_{13}(x_1, x_3) = \begin{cases} (1 - e^{-x_1})(1 - e^{-3x_3}) & \text{for } x_1, x_3 \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

◀

Example: Consider that three continuous RVs X, Y and Z are jointly uniformly distributed in the region $0 \leq z \leq y \leq x \leq 1$.

1. Find the multivariate joint PDF $f_{XYZ}(x, y, z)$.
2. Find the marginal joint PDFs $f_{XY}(x, y)$, $f_{XZ}(x, z)$ and $f_{YZ}(y, z)$.
3. Find the marginal PDFs $f_X(x)$, $f_Y(y)$ and $f_Z(z)$,

Since $f_{XYZ}(x, y, z)$ is jointly continuous, $f_{XYZ}(x, y, z) = c$

Multivariate joint PDF

$$\int_0^1 \int_0^x \int_0^y c \, dz \, dy \, dx = 1$$

$$= \int_0^1 \int_0^x \left(cz|_0^y \right) \, dy \, dx = \int_0^1 \frac{cy^2}{2}\bigg|_0^x \, dx$$

$$= \int_0^1 \frac{1}{2} c x^2 \, dx = \frac{1}{2} c \frac{x^2}{3}\bigg|_0^1 = \frac{c}{6}$$

$$\Rightarrow c = 6$$

Marginal joint PDF of any pair of X, Y and Z

$0 \leq z \leq y \leq x \leq 1$			
RVS	z	y	x
Limits	$0 \leq z \leq y$	$z \leq y \leq x$	$y \leq x \leq 1$

Marginal Joint PDF $f_{XY}(x,y)$		
PDF	$f_{XY}(x,y) = \int_0^y 6 \; dz$ $= 6z\|_0^y$ $= 6y$	$f_{XY}(x,y) = \begin{cases} 6y, & 0 \le y \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$
Proof	$\int_0^1 \int_0^x 6y \; dy \, dx = \int_0^1 \left(6 \frac{y^2}{2} \Big\|_0^x \right) dx = \int_0^1 3x^2 \, dx = 3 \frac{x^3}{3} \Big\|_0^1 = 1$	

Marginal Joint PDF $f_{XZ}(x,z)$		
PDF	$f_{XZ}(x,z) = \int_z^x 6\ dy$ $= 6y\vert_z^x$ $= 6(x-z)$	$f_{XZ}(x,z) = \begin{cases} 6(x-z), & 0 \le z \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$
Proof	$\int_0^1 \int_0^x 6(x-z)\ dz\, dx = \int_0^1 \left(6xz\vert_0^x - 6\frac{z^2}{2}\bigg\vert_0^x \right) dx = \int_0^1 (6x^2 - 3x^2)\, dx$ $= 3\frac{x^3}{3}\bigg\vert_0^1 = 1$	

Marginal Joint PDF $f_{YZ}(y,z)$		
PDF	$f_{YZ}(y,z) = \int_y^1 6 \ dx$ $= 6x\vert_y^1$ $= 6(1-y)$	$f_{YZ}(y,z) = \begin{cases} 6(1-y), & 0 \le z \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$
Proof	$\int_0^1 \int_0^y 6(1-y) \ dz \, dy = \int_0^1 \left(6z\vert_0^y - 6yz\vert_0^y\right) dy = \int_0^1 (6y - 6y^2) \, dy$ $= 6\frac{y^2}{2}\Big\vert_0^1 - 6\frac{y^3}{3}\Big\vert_0^1 = 3 - 2 = 1$	

<table>
<tr><th>Marginal PDF of X</th><th>Marginal joint PDF of X,Y and X,Z</th></tr>
<tr><td>$$f_X(x) = \int_0^x 6y\,dy = 6\frac{y^2}{2}\bigg|_0^x = 3x^2$$</td><td>$$f_{XY}(x,y) = \begin{cases} 6y, & 0 \le y \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$</td></tr>
<tr><td>$$f_X(x) = \int_0^x 6(x-z)\,dz$$ $$= 6xz|_0^x - 6\frac{z^2}{2}\bigg|_0^x = 6x^2 - 3x^2$$ $$= 3x^2$$</td><td>$$f_{XZ}(x,z) = \begin{cases} 6(x-z), & 0 \le z \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$</td></tr>
<tr><td colspan="2">$$f_X(x) = \begin{cases} 3x^2, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$</td></tr>
</table>

Marginal PDF of Y	Marginal joint PDF of X, Y and Y, Z
$f_Y(y) = \int_y^1 6y\,dx = 6yx\rvert_y^1 =$ $= 6y - 6y^2 = 6y(1-y)$	$f_{XY}(x,y) = \begin{cases} 6y, & 0 \le y \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$
$f_Y(y) = \int_0^y 6(1-y)\,dz$ $= 6z\rvert_0^y - 6yz\rvert_0^y = 6y - 6y^2$ $= 6y(1-y)$	$f_{YZ}(y,z) = \begin{cases} 6(1-y), & 0 \le z \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$
$f_Y(y) = \begin{cases} 6y(1-y), & 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$	

Marginal PDF of Z	Marginal joint PDF of X, Z and Y, Z
$f_Y(y) = \int_y^1 6y\,dx = 6yx\rvert_y^1 =$ $= 6y - 6y^2 = 6y(1-y)$	$f_{XY}(x,y) = \begin{cases} 6y, & 0 \le y \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$
$f_Y(y) = \int_0^y 6(1-y)\,dz$ $= 6z\rvert_0^y - 6yz\rvert_0^y = 6y - 6y^2$ $= 6y(1-y)$	$f_{YZ}(y,z) = \begin{cases} 6(1-y), & 0 \le z \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$
$f_Y(y) = \begin{cases} 6y(1-y), & 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$	

Random Variables $Y_1, \dots, Y_4$ have the joint PDF

$$f_{Y_1,\dots,Y_4}(y_1,\dots,y_4) = \begin{cases} 4 & 0 \le y_1 \le y_2 \le 1, 0 \le y_3 \le y_4 \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal PDFs $f_{Y_1,Y_4}(y_1, y_4)$, $f_{Y_2,Y_3}(y_2, y_3)$ and $f_{Y_3}(y_3)$.

For $0 \le y_1 \le 1$ and $0 \le y_4 \le 1$

$$f_{Y_1,Y_4}(y_1,y_4) = \int_{y_1}^{1} \int_{0}^{y_4} 4\, dy_3\, dy_2 = 4(1-y_1)y_4.$$

The complete expression for $f_{Y_1,Y_4}(y_1, y_4)$ is

$$f_{Y_1,Y_4}(y_1, y_4) = \begin{cases} 4(1 - y_1)y_4 & 0 \leq y_1 \leq 1, 0 \leq y_4 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Similarly, for $0 \leq y_2 \leq 1$ and $0 \leq y_3 \leq 1$,

$$f_{Y_2,Y_3}(y_2, y_3) = \int_0^{y_2} \int_{y_3}^{1} 4\, dy_4\, dy_1 = 4y_2(1 - y_3).$$

The complete expression for $f_{Y_2,Y_3}(y_2, y_3)$ is

$$f_{Y_2,Y_3}(y_2, y_3) = \begin{cases} 4y_2(1 - y_3) & 0 \leq y_2 \leq 1, 0 \leq y_3 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

For $0 \leq y_3 \leq 1$

$$f_{Y_3}(y_3) = \int_{-\infty}^{\infty} f_{Y_2,Y_3}(y_2, y_3)\, dy_2 = \int_0^1 4y_2(1 - y_3)\, dy_2 = 2(1 - y_3).$$

The complete expression is

$$f_{Y_3}(y_3) = \begin{cases} 2(1 - y_3) & 0 \leq y_3 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$